Tower Bridge - Grown-ups

UK Family Tour

Tower Bridge: Victorian Engineering in Gothic Costume

Tower Bridge is a piece of misdirection: it looks medieval, but it's a steel bridge of the 1890s wearing granite and Portland stone as cladding. The costume was a planning requirement — Parliament insisted the new bridge harmonize with the Tower of London next door — but underneath is over eleven thousand tonnes of steel and what was, at opening in 1894, the most sophisticated bascule mechanism in the world. "Bascule" is French for see-saw: each thousand-tonne leaf pivots on counterweights so perfectly balanced that the original hydraulic system, powered by coal-fired steam engines pumping water to enormous accumulators, could raise the road in about a minute. The problem it solved was real: by the 1880s the East End had a million people and one crossing at London Bridge, but a fixed bridge would have strangled the busiest docks on Earth. The bascule design let both be true. The bridge still opens around eight hundred times a year, electric since 1976, and river traffic retains absolute priority — bookings are free with twenty-four hours' notice, and road traffic waits, whoever it is. The Engine Rooms preserve the original steam plant, and it's worth the time: this is the machinery of the age that assumed any problem could be solved with enough brass, coal and confidence.

> Related listening: Two great Victorian bridges, opened four years apart: one dressed as a castle, one all bare steel. Coming up — "The Forth Bridge: Engineering as Public Reassurance" (Inchcolm Island & Forth Bridge, Day 10).

The Power in the Cellar

The genius of Tower Bridge was never the see-saw idea alone. Balance a thousand-tonne leaf on a counterweight and you still need to move it, on demand, in about a minute, at any hour of the day or night, whether or not you happen to have a full head of steam at that moment. The Victorians solved this with a trick that feels almost magical once you see it, and the whole apparatus survives, gleaming, in the Engine Rooms downstairs. It is the best part of the visit, and most people rush past it.

Here is the logic. Coal-fired boilers raised steam, and the steam drove great pumping engines, but those pumps did not power the bridge directly. Instead they forced water, under enormous pressure, into devices called accumulators. Picture a tall iron column with a colossal weighted ram riding on top of a trapped column of water. The pumps squeeze water in, the huge weight is pushed upward, and there it sits, patiently pressing down, holding all that force in reserve like a wound spring or a charged battery made of pressure rather than electricity. When a keeper called for a lift, that stored pressure was released to the driving engines in an instant, and the leaves rose whether or not the boilers were at full heat. The bridge always had power ready and waiting.

This was cutting-edge for its time, part of a great age of hydraulic power that quietly ran Victorian Britain, working the cranes of the docks, the lifts of grand hotels, and the machinery of theatres. Water at pressure was the city's hidden muscle, and Tower Bridge is one of its finest surviving cathedrals.

The steam era ended in nineteen seventy-six, when the bridge was converted to a modern oil and electric system. The change was sensible, cleaner and far cheaper to run, and it retired the men who had shovelled coal for eighty years. But the original engines were not scrapped. They were preserved in place, polished to their Sunday best in green and red, and you can walk among them today, along with one of the surviving accumulators. Stand beside those machines and you are looking at the confident heart of the nineteenth century, an age that assumed any problem at all could be beaten with enough coal, brass, and nerve.

How a Thousand Tonnes Balances

It is worth pausing on the sheer physical audacity of what happens when Tower Bridge opens, because the numbers are genuinely hard to believe. Each of the two moving leaves, counterweight and all, weighs well over a thousand tonnes. That is the mass of a small warship, hinged at one end, swinging up into the air, and then set down again to meet its partner across a gap with the precision of a cabinetmaker closing a drawer. And it does this hundreds of times a year without drama.

The word bascule is simply French for see-saw, and the principle really is that homely. Most of each leaf's weight is not the roadway you walk on but a vast counterweight, buried out of sight in the pier below the road, on the short side of the pivot. The leaf turns on massive steel shafts called trunnions, the axle of the whole see-saw, and it is balanced so finely that the great counterweight does nearly all the lifting for you. The machinery does not really heave the road skyward through brute force. It nudges an object that is already teetering on the edge of moving, then guides it, tooth by tooth, as engines drive pinions around curved toothed racks fixed to the leaf.

Because the balance is so close to perfect, the energy required is astonishingly modest for the scale of the thing. The leaves rise to a steep angle, close to vertical, opening a clear channel for tall vessels to pass. The actual lift takes only about a minute, though the full sequence, stopping traffic, closing barriers, raising, waiting for the ship, lowering and locking, runs to several minutes from start to finish.

There is a lovely lesson buried in this for anyone who admires clever design. The Victorians did not conquer the weight of the bridge. They befriended it. Rather than fighting a thousand tonnes with a thousand tonnes of force, they arranged for the bridge to be forever on the verge of moving by itself, held in check, needing only permission to swing. When you next watch those leaves climb toward the sky with such unhurried ease, remember that you are not watching muscle. You are watching balance, the oldest trick in engineering, performed at a scale almost nobody else has ever dared to attempt.

Building in a Living River

It is easy to admire the finished bridge and forget the sheer difficulty of putting it there. Consider the problem the builders faced in the eighteen eighties. They had to sink two enormous piers into the bed of the Thames, in the middle of the busiest working river on the planet, and do it without ever stopping the ceaseless traffic of ships that was the whole reason the bridge existed. You cannot dam the Thames. You cannot ask the port of London to take a few years off. The construction had to happen around the river, not instead of it.

The solution was to work inside great watertight enclosures. Massive iron caissons and cofferdams were sunk to the riverbed and pumped dry, creating pockets of open air far below the waterline where men could lay the foundations in the muck, while the tide rose and fell and ships passed close by overhead. Into those foundations went the two piers, and onto the piers rose the steel skeleton, over eleven thousand tonnes of it, the true structure of the bridge. Only then came the disguise, the granite and pale Portland stone quarried and hauled to the site and hung on the steel like a mason's mask, so that the finished machine could pass for a medieval gatehouse and satisfy the planners who insisted it defer to the Tower next door.

The whole undertaking took the better part of a decade and consumed a small fortune, and it was dangerous, patient, unglamorous work, carried out by hundreds of labourers, masons, and divers in cold water and river mud. There is a tendency to picture the Victorians simply willing their monuments into existence through confidence alone. The truth beneath this one is more impressive, a grind of tides, pumps, and heavy stone, executed with a precision that has held the roadway level for well over a century.

So when you cross the bridge or gaze up at those towers, spare a thought for what lies out of sight, below the waterline you cannot see. The elegant Gothic show above rests on a hidden feat of stubborn, muddy engineering, two islands of masonry driven into a tidal river that never once paused to let the builders work. The costume gets the attention. The foundations earned it.

The High Walkways: A Social History in Glass

The high-level walkways, forty-two meters above the Thames, were part of the original design's logic: when the bascules rose, pedestrians could climb the towers and keep crossing. In practice, Victorians voted with their feet — waiting a few minutes to watch a ship pass was more appealing than climbing two hundred stairs — and the underused walkways developed a reputation as a haunt for pickpockets and prostitution. They closed in 1910 and stayed shut for over seventy years, reopening in 1982 as part of the exhibition, with the glass floor sections added in 2014. Each glass panel weighs over half a tonne, five laminated layers rated far beyond any plausible load. The view is one of the best in London for understanding the city's geography: upstream, the Shard and the City's towers; downstream, Canary Wharf, built on the very docks whose ship traffic justified this bridge. And the 1952 bus incident is well documented: driver Albert Gunter accelerated his number 78 double-decker across the opening gap after a watchman's error let traffic on during a lift, clearing roughly a meter's drop to the north bascule. No serious injuries; Gunter received a reward of ten pounds. The gate procedures were, unsurprisingly, revised.

Vice, Closure, and Resurrection

The high walkways are one of the great redemption stories in London's built history, an idea that failed, fell into disgrace, and came back transformed. It is worth knowing the arc while you stand up here, because the elegant glass gallery around you was, for the better part of a century, a place respectable people were warned to avoid.

The original reasoning was sound on paper. When the bascules rose for a ship, foot passengers need not wait. They could ascend the towers, cross the river high above the open road, and descend the far side, keeping the flow of pedestrians unbroken. The Victorians, however, declined to cooperate with the logic. Climbing some two hundred steps to save a few minutes held little appeal when one could simply stand at the roadside and enjoy the free spectacle of a great ship passing. The walkways went almost unused, and an unwatched, elevated, semi-enclosed corridor above a teeming city acquired precisely the reputation you would expect. They became a resort of pickpockets, and were long associated with prostitution, a discreet high place beyond the reach of the constable on the ground. In nineteen ten the authorities gave up and closed them.

Then came the long silence. For over seventy years the walkways stood shut and neglected, sealed away above the traffic while two world wars and the whole convulsion of the twentieth century unfolded below. They were, in effect, a forgotten room in one of the most photographed structures on Earth.

Their resurrection came in nineteen eighty-two, when the walkways were reopened as part of a new exhibition, the paid attraction that keeps the bridge earning its living to this day. The final flourish arrived in twenty fourteen, when sections of the floor were replaced with glass, turning a Victorian afterthought into a genuine thrill and one of the most Instagrammed vantage points in the capital.

There is a neat irony in all of this. The feature the Victorians built for pure utility and then abandoned as useless and disreputable has become, in our age, the very reason many visitors climb the towers at all. What could not persuade a nineteenth-century pedestrian to take two hundred stairs now draws crowds who queue for the privilege. The walkways failed at their job for a hundred years, then found a far better one. Few pieces of infrastructure get a second act this complete.

The Neighbours from the Walkway

The walkway panorama is usually praised for its towers of glass and money, the Shard, the City, Canary Wharf. But turn your attention to the older, darker neighbours, because from up here you can read a thousand years of English power and cruelty in a single sweep.

Begin with the fortress right beside you, the Tower of London, which predates this bridge by more than eight centuries. Its oldest core, the White Tower, was raised by William the Conqueror after ten sixty-six, as much to overawe the citizens of London as to defend them. Look for the waterline along the river frontage and you may pick out Traitors' Gate, the arched river entrance through which prisoners were brought in by boat, under the wharf, to be swallowed by the fortress. To arrive at the Tower by water was very often to leave it only for the scaffold. Anne Boleyn, Thomas More, Lady Jane Grey, and a long procession of the once-powerful passed this way. The bridge you are standing on was deliberately dressed in mock-medieval Gothic to defer to that grim old building, the modern machine bowing to the ancient fortress.

Now find the grey warship moored in the river, all guns and superstructure. That is HMS Belfast, a Royal Navy cruiser that served through the Second World War, including the Arctic convoys and the bombardment of the Normandy beaches on D-Day. She is a museum now, but her main guns were real enough, capable of hurling a shell many miles inland, and a favourite piece of London trivia claims they are trained on a motorway service station somewhere out past the northern suburbs. It is a joke about their range, not a genuine threat, but it lands because the range was real.

Between these landmarks flows the working river itself, the reason for everything you can see. This narrow stretch was once the beating heart of the greatest port on the planet, and the Tower stood guard over its wealth while the bridge below you let the ships come and go.

So take a slow turn and read the layers. A conqueror's fortress and its gate of no return, a warship that shelled a continent, and, holding it all together, a Victorian drawbridge in fancy dress. Few balconies anywhere let you take in quite so much history, and quite so much of it soaked in blood and salt water, without moving your feet.

Incidents and Iconography

Tower Bridge's afterlife as a symbol has been eventful. The 1952 bus jump is the canonical story — driver Albert Gunter accelerating his number 78 across the rising bascule after a relief watchman failed to close the gates, clearing about a meter's drop, injuries minor, reward ten pounds — but it has company. In 1968, an RAF pilot, aggrieved over jubilee flypast plans, flew a Hawker Hunter jet through the span between the walkways and the road, unauthorized, at several hundred miles an hour; he was arrested on landing and quietly retired rather than court-martialed, public sympathy being firmly with the pilot. In 1997, President Clinton's motorcade was split in two when the bridge opened, exactly on schedule, for a Thames barge — the harbormaster's office noting, in effect, that the river had booked first. Even the paint tells a story: the bridge was chocolate brown for much of the twentieth century and acquired its patriotic blue, white and red for the Queen's Silver Jubilee in 1977. And a persistent legend claims an American bought the 1960s' demolished London Bridge believing it was this one; the buyer always denied it, but the story refuses to die because it flatters a truth — Tower Bridge is the bridge the world pictures when it thinks of London, a Victorian machine that became a logo. It remains a working crossing: forty thousand people a day, and the river still outranks them all.

Aviators and Daredevils

Tower Bridge has a peculiar magnetism for pilots. That narrow slot between the raised walkways and the roadway, roughly forty-three meters of clear air, has proved an irresistible dare for well over a century, and two flights in particular have entered legend.

The first came astonishingly early. On the tenth of August, nineteen twelve, when powered flight was barely older than a toddler and most Londoners had never seen an aircraft at all, Frank McClean flew a Short biplane fitted with floats up the Thames. Reaching Tower Bridge, he threaded the machine between the high walkways and the road, straight through the gap, then continued upriver, flying beneath the succeeding bridges and skimming just above the water before coming ashore near Westminster. In a flimsy contraption of wood, wire, and canvas, with no margin for error and no second attempt, it was an act of quite extraordinary nerve. McClean later cast it as a protest, a pointed demonstration to a government he felt was failing to take aviation seriously. He was, by any measure, decades ahead of his time.

The second flight was faster, louder, and far more mischievous. On the fifth of April, nineteen sixty-eight, Flight Lieutenant Alan Pollock of the Royal Air Force was incensed that the top brass had planned no display to mark the fiftieth anniversary of the RAF. So, entirely on his own initiative and without a shred of authorisation, he broke away from his flight, beat up several London landmarks at low level in his Hawker Hunter, and then flew the jet clean through the span of Tower Bridge at several hundred miles an hour. Others had flown through the bridge before him, but Pollock was the first ever to do it in a jet. He was arrested on landing and, rather than face a court martial, was quietly invalided out of the service on medical grounds. Public sympathy sat firmly with the pilot, and he remained a folk hero until his death in twenty twenty-five, at the age of eighty-nine.

Two men, fifty-six years apart, one in a floatplane held together with string and one in a fighter jet, both drawn to the same slender opening between the towers. The bridge, it seems, does not merely span the river. It issues a standing challenge to anyone bold or foolish enough to accept it, and every so often, someone still does.

The Great Bridge Mix-Up

No account of Tower Bridge is complete without the most delicious case of mistaken identity in the city's history, one that ends, improbably, in the Arizona desert.

The confusion has deep roots. Foreign visitors have muddled Tower Bridge with London Bridge for as long as both have existed, and the muddle produced one of the world's favourite anecdotes. In nineteen sixty-eight the City of London decided to sell off its London Bridge, not the famous medieval one crowded with houses, but its sober Victorian successor, the granite crossing designed by John Rennie and opened in eighteen thirty-one, which was slowly sinking under the weight of modern traffic. An American entrepreneur named Robert McCulloch bought it, for something close to two and a half million dollars, and had it dismantled stone by numbered stone, shipped across the Atlantic and the American continent, and painstakingly reassembled beside a reservoir at Lake Havasu City, a brand-new town he was building in the Arizona desert. It was rededicated there in nineteen seventy-one, and it draws tourists to this day, a genuine piece of old London marooned in the sagebrush.

Almost immediately, a irresistible story attached itself to the sale. McCulloch, it was whispered, had blundered. He had believed he was buying the far grander, turreted Tower Bridge, the one the whole world pictures, and had been horrified to unwrap a plain grey road bridge instead. It is a wonderful tale. It is also, by every reliable account, untrue. Ivan Luckin, the City councillor who brokered the sale, was asked directly whether McCulloch had thought he was getting Tower Bridge and replied simply, "Of course not." McCulloch's own family have insisted for decades that he knew exactly what he was purchasing. The legend survives not because it is true but because it flatters a real fact, that Tower Bridge is so overwhelmingly the image people carry of London that we assume any buyer must have wanted it.

And there lies the quiet triumph of the bridge you are standing on. It became such a perfect symbol of the city that a myth grew up in which a rich man crossed an ocean to buy it by mistake. The plainer London Bridge got the sale and the desert exile. Tower Bridge got something better. It got to be the bridge that everyone, everywhere, means when they say London.